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C1 — Particles

OCR GCSE Chemistry A (J248) — Gateway Science · Papers 1 & 3 (C1.1 The particle model; C1.2 Atomic structure)

Topic C1 is the foundation for everything else in the course. It sets out the **particle model** of solids, liquids and gases, and then looks inside the particles to describe **atomic structure**: protons, neutrons, electrons, isotopes and ions. Topic C1 is assessed on Paper 1 (Foundation) and Paper 3 (Higher), and the ideas are assumed knowledge on Papers 2 and 4.

Content marked **(HT)** is tested only on the Higher tier papers. Everything else can be tested on both tiers.

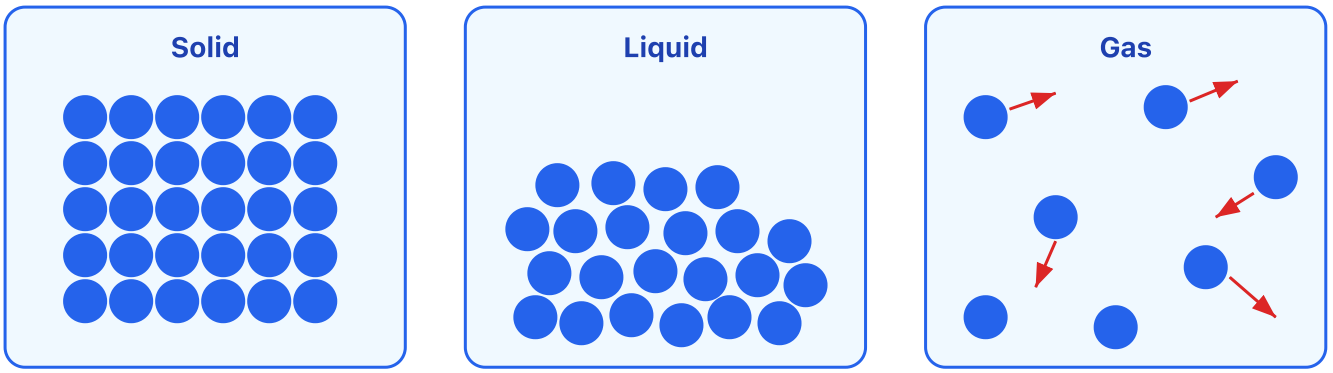
C1.1 The Particle Model

> The **particle model** says that all matter is made of very small particles (atoms, molecules or ions) that are **constantly moving**. Whether a substance is a solid, liquid or gas depends on how its particles are **arranged**, how they **move** and how strongly they are **held together**.

The model is a *model*: a simplified picture used to explain and predict observations. In the simple version the particles are drawn as small, solid spheres. That is enough to explain melting, boiling, diffusion and compression, but (as you will see under the limitations below) it leaves a lot out.

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Particle Arrangement in Solids, Liquids and Gases



Regular lattice · touching
vibrate about fixed positions

Random · mostly touching
move around each other

Random · far apart
move quickly in all directions

Properties of the Three States

Feature	Solid	Liquid	Gas
Arrangement	Regular, repeating pattern (lattice)	Random, but particles close together	Random, particles far apart
Movement	Vibrate about fixed positions	Move around / slide past each other	Move quickly and randomly in all directions
Forces between particles	Strong	Weaker than in solid	Very weak (almost none)
Energy of particles	Lowest	More	Most
Shape	Fixed	Takes the shape of the container	Fills the whole container
Volume	Fixed	Fixed	Not fixed — expands to fill space
Can it be compressed?	No	Hardly at all	Yes — large spaces between particles

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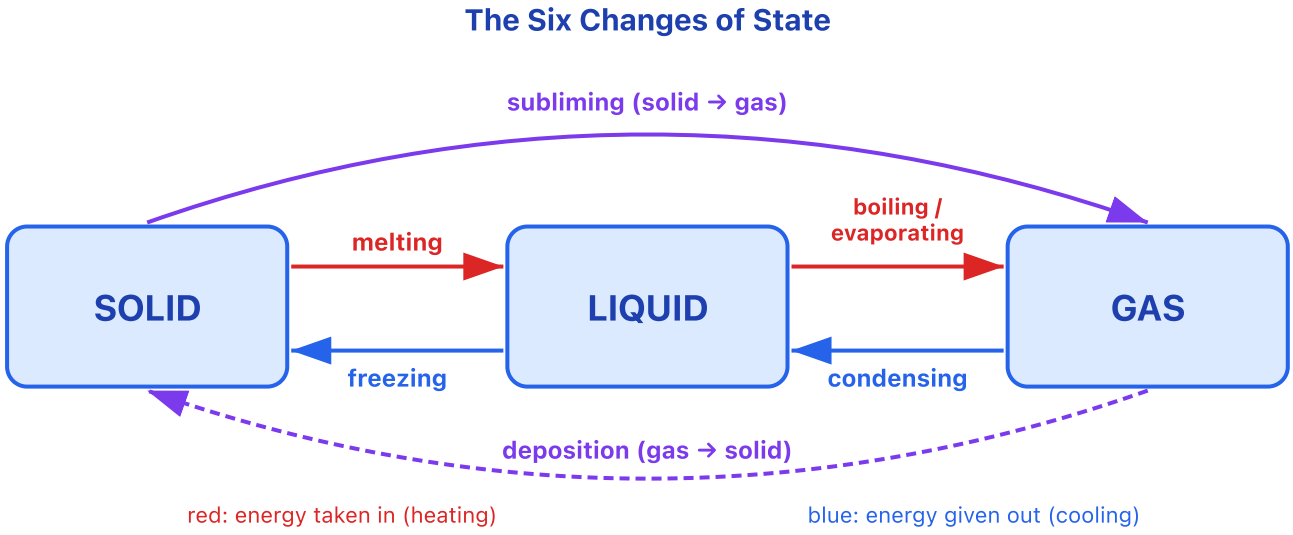
Can it flow?	No	Yes	Yes

The particle model explains these bulk properties. A gas can be squashed because there is lots of empty space between its particles. A liquid flows because its particles can move past one another. A solid keeps its shape because its particles are held in fixed positions by strong forces.

> **Exam Tip:** When drawing particle diagrams, keep the particles the **same size** in all three states. In a liquid, most particles should still be touching (no big gaps) — a very common mistake is to draw a liquid that looks like a gas. Particles do not get bigger when heated; they just move faster and (usually) further apart.

Changes of State

A **change of state** happens when energy is transferred to or from a substance. Heating gives the particles more kinetic energy, so they can overcome more of the forces holding them together. Cooling removes energy, so forces of attraction pull the particles closer together.



Change	From → to	Energy	What happens to the particles
Melting	solid → liquid	taken in	Particles vibrate so much that they break free of fixed positions; enough forces are overcome for them to move around each other.

Freezing	liquid → solid	given out	Particles slow down; forces of attraction hold them in fixed positions.
Boiling / evaporating	liquid → gas	taken in	Particles gain enough energy to overcome almost all the forces and escape. Boiling happens throughout the liquid at the boiling point; evaporation happens only from the surface, at any temperature.
Condensing	gas → liquid	given out	Particles slow down and forces pull them close together again.
Subliming	solid → gas	taken in	Particles go straight from fixed positions to moving freely, e.g. solid carbon dioxide (dry ice), iodine.

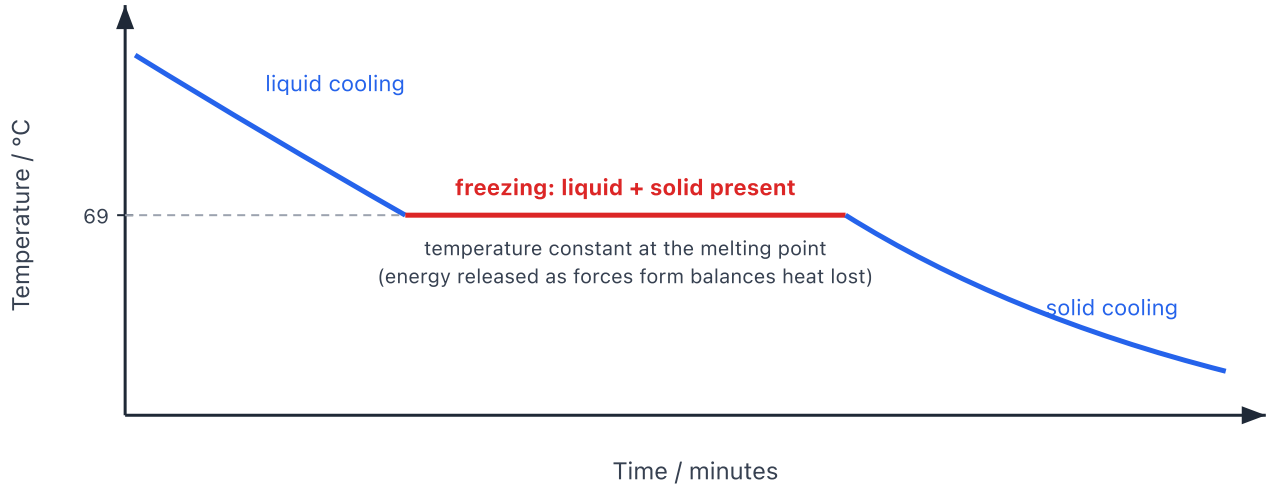
Changes of state are **physical changes**: no new substance forms, the particles themselves are unchanged and the change is **reversible**. The **mass is conserved** because the number of particles stays the same — they are only rearranged. (Ice that melts in a sealed flask has the same mass as the water it becomes.)

Heating and Cooling Curves

If a pure substance is heated steadily, its temperature rises until it reaches its melting point. The temperature then **stays constant** while the substance melts, even though energy is still being supplied, because the energy is being used to overcome the forces between particles rather than to make them move faster. The same happens at the boiling point.

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Cooling Curve for a Pure Substance (e.g. stearic acid)



The flat section on a cooling curve is where the substance freezes. Its temperature tells you the **melting point** (melting point = freezing point for a pure substance). A *pure* substance gives a sharp, flat plateau; an impure one solidifies over a range of temperatures (see C2.1).

Melting Point, Boiling Point and State at a Given Temperature

- Below its melting point a substance is a **solid**.
- Between its melting point and boiling point it is a **liquid**.
- Above its boiling point it is a **gas**.

The *size* of the melting and boiling points depends on how strong the forces between particles are. Stronger forces need more energy to overcome, so the melting point is higher. Topic C2.3 explains this in terms of bonds and intermolecular forces.

Diffusion and Gas Pressure (Using the Model)

The particle model also explains:

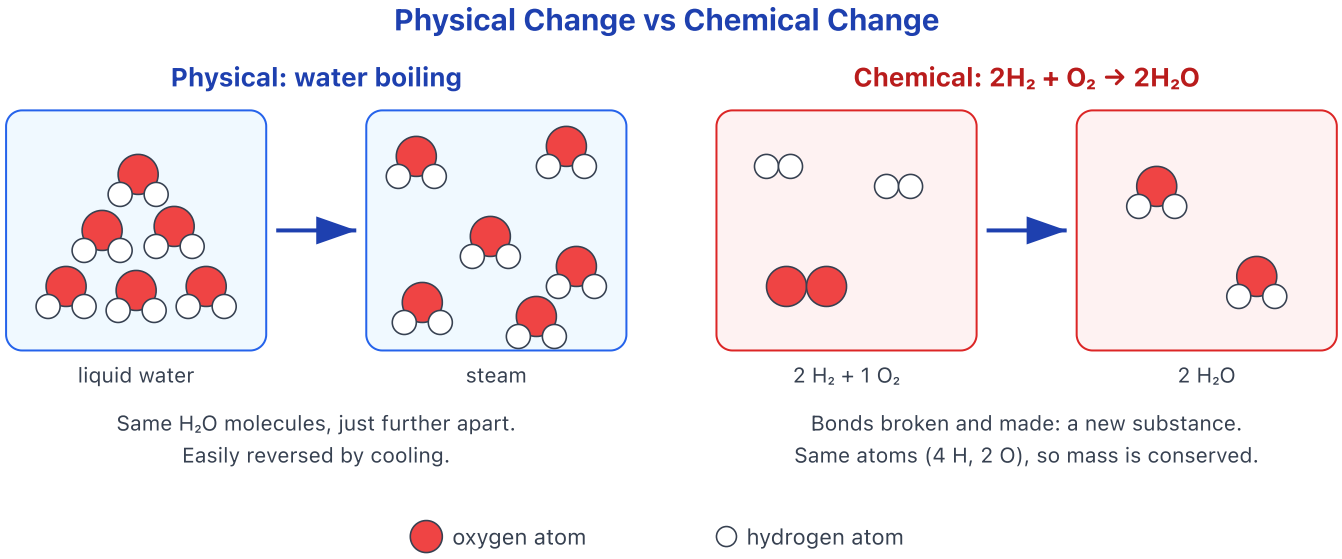
- **Diffusion** — the net movement of particles from an area of higher concentration to lower concentration because of their random motion. Smells spread across a room; a purple crystal of potassium manganate(VII) slowly colours still water. Diffusion is fastest in gases (fast particles, large spaces) and very slow in liquids.
- **Gas pressure** — gas particles collide with the walls of their container. Heating the gas makes the particles move faster, so they collide more often and harder — the pressure increases.

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Physical Changes and Chemical Changes

The particle model gives a clear way to tell physical and chemical changes apart:

> In a **physical change** the particles stay the same — they are only rearranged or moved further apart / closer together. No new substance forms. Examples: melting, boiling, dissolving.
>
> In a **chemical change** (a chemical reaction) the **atoms are rearranged into new substances**: bonds between atoms are broken and new bonds are made. Examples: burning, rusting, neutralisation.



	Physical change	Chemical change
New substance formed?	No	Yes
What happens to particles	Same particles; spacing / arrangement / movement changes	Atoms rearranged; bonds broken and new bonds formed

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Reversible?	Usually easily reversed (e.g. cool steam → water)	Usually hard to reverse
Energy change	Usually small	Often large (heat, light, sound)
Mass	Conserved	Conserved (in a closed system)
Signs	Change of state, shape or dissolving only	Colour change, gas given off (fizzing), precipitate, temperature change, light
Examples	Ice melting; sugar dissolving; distilling salty water	Magnesium burning; iron rusting; baking a cake

Worked Example: When magnesium ribbon is heated in air it burns with a bright white light and becomes a white powder. Is this a physical or chemical change? Explain using the particle model.
Chemical change. The magnesium atoms react with oxygen molecules from the air: the bonds in O_2 are broken and new bonds between magnesium and oxygen form, giving a **new substance**, magnesium oxide ($2Mg + O_2 \rightarrow 2MgO$). The white powder cannot be turned back into magnesium by cooling. Compare this with melting wax, where the same wax molecules simply move further apart and the wax re-solidifies on cooling.

Exam Tip: Dissolving is a **physical** change, not a chemical reaction: salt can be recovered by evaporating the water. Don't say the salt "disappears" — its particles spread out between the water particles, and the total mass of the solution = mass of salt + mass of water.

Elements, Compounds and Mixtures (Particle Diagrams)

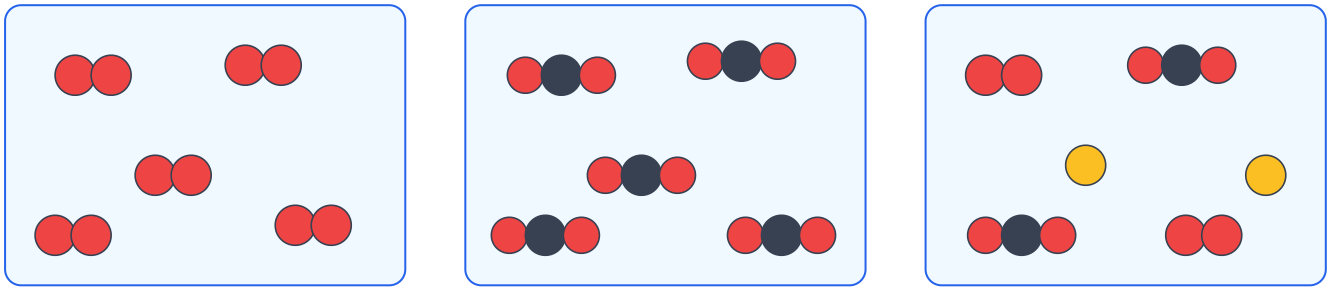
The particle model on its own does not explain why different materials have different properties — to do that we need to look at the particles themselves. Different substances are made of different particles:

- An **element** contains only **one type of atom** (e.g. Fe, O_2, He). It cannot be broken down into simpler substances by chemical means.

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- A **compound** contains atoms of **two or more different elements chemically bonded** in fixed proportions (e.g. H_2O, NaCl, CO_2).
- A **mixture** contains two or more substances **not chemically joined**; they can be separated by physical methods (C2.1).

Particle Diagrams: Element, Compound, Mixture



Element (oxygen, O_2)
one type of atom

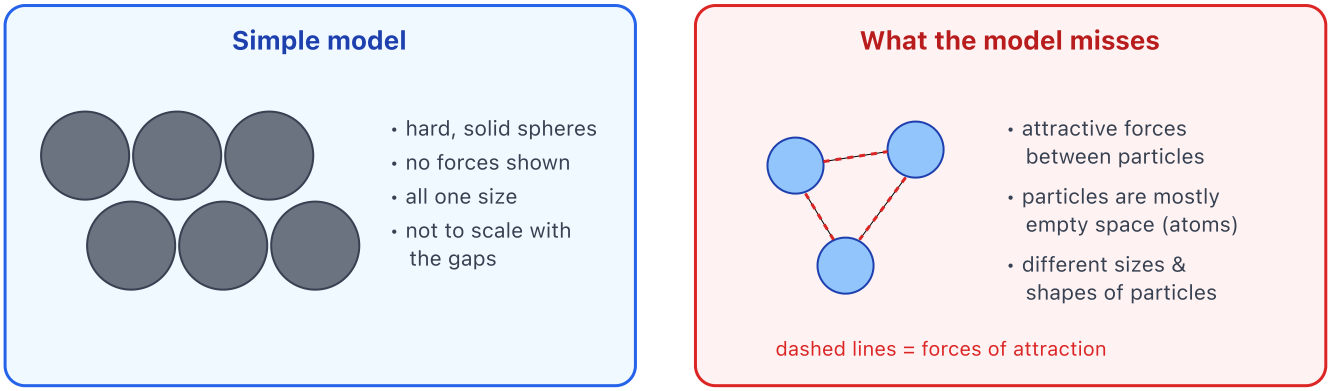
Compound (carbon dioxide, CO_2)
different atoms bonded together

Mixture (O_2 + CO_2 + He)
substances not bonded to each other

Limitations of the Particle Model (HT)

In the simple particle model, particles are drawn as **small, solid, inelastic spheres** — like bowling balls. This is useful, but it is not how real particles behave. You need to be able to explain the limitations of this model when it is used to explain **changes of state**.

(HT) "Bowling Ball" Model vs Real Particles



The model shows **where** particles are, but not **why** they stay together or how much energy is needed to separate them — so it cannot predict melting or boiling points.

Limitation (HT)	Why it matters for changes of state

It does **not take into account the forces of attraction** between particles	Changes of state happen when these forces are overcome (melting, boiling) or formed (condensing, freezing). Without forces, the model cannot explain why energy is needed to melt or boil a substance, or why different substances melt at different temperatures.
It does **not take into account the size of the particles**	Real particles differ in size and shape (a helium atom vs a large wax molecule), which affects how strongly they attract each other and so their melting / boiling points. The spheres are all drawn the same size.
It does **not take into account the space between the particles**	Diagrams are not to scale. In a gas the particles are much further apart than shown (about 10 particle diameters apart at room temperature), and particles themselves are mostly empty space, not solid.
Particles are treated as **inelastic** (solid, hard) spheres	Real particles are not solid balls; they do not have fixed edges and can be distorted. Molecules can also vibrate and rotate.
Particles are shown **still** in a diagram	Real particles are always moving (even in a solid they vibrate), and the amount of movement depends on temperature.

Exam Tip: (HT) The three limitations named in the specification are: the model ignores the **forces of attraction between particles**, the **size of the particles**, and the **space between them**. In a 2- or 3-mark question, name the limitation *and* link it to the change of state, e.g. "the model doesn't show the forces between particles, so it can't explain why energy has to be supplied to melt the solid".

Practical opportunity (C1.1): observe changes of state and compare them with chemical changes — e.g. heat ice in a beaker and record the temperature every 30 s to plot a heating curve; melt and re-solidify wax or stearic acid (physical, reversible) and compare with heating magnesium ribbon or copper carbonate (chemical: new substance, not reversed on cooling). There is no assessed PAG attached to Topic C1, but this work develops the measuring skills used in PAG C4 and PAG C8.

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C1.2 Atomic Structure

An **atom** is the smallest part of an element that still has the chemical properties of that element. Atoms are made of three **sub-atomic particles**: **protons** and **neutrons** in a tiny central **nucleus**, and **electrons** arranged in shells (energy levels) around it.

How the Model of the Atom Has Changed

Our model of the atom has been developed over 200 years. Each time new **experimental evidence** appeared that the old model could not explain, the model was changed or replaced. This is a key example of how science works: models are tested, and kept only while they fit the evidence.

Timeline of the Atomic Model



Each change was driven by new evidence the previous model could not explain.

Scientist(s)	Model / discovery	Evidence and reason for change
John Dalton (early 1800s)	Atoms are tiny, solid, indivisible spheres. Atoms of the same element are identical; different elements have different atoms. Atoms combine in simple whole-number ratios to make compounds.	Based on the fixed ratios in which elements combine (laws of chemical combination). First modern atomic theory.

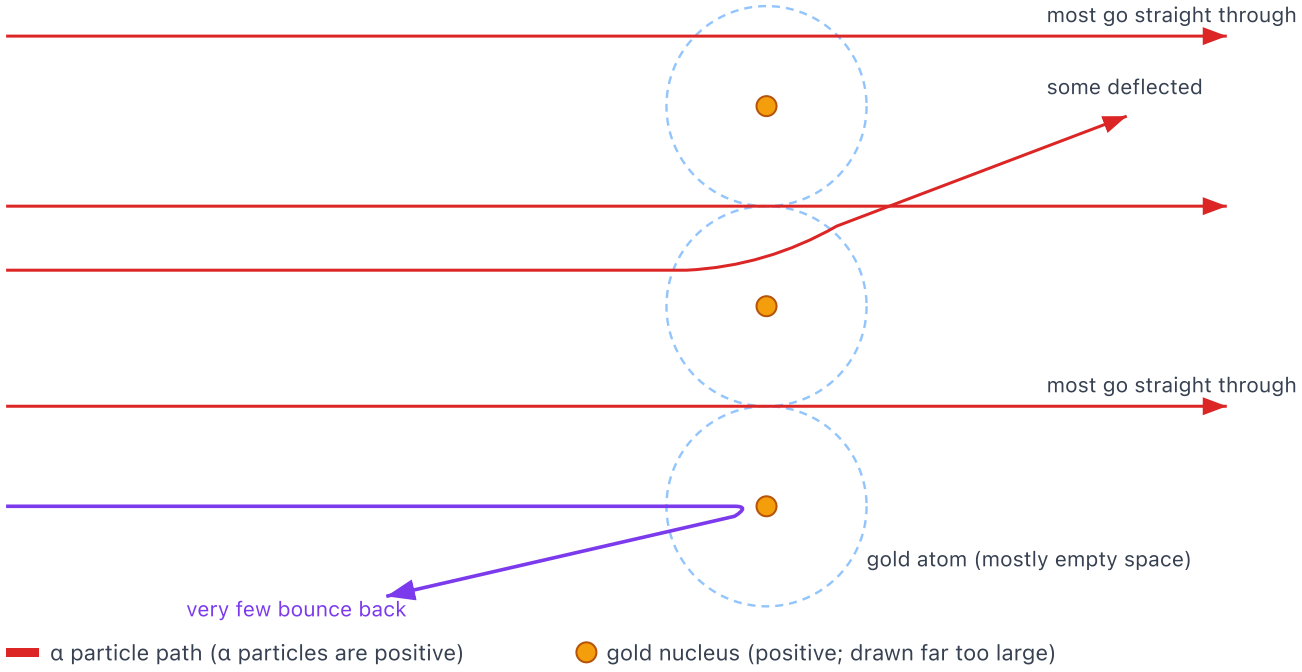
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J. J. Thomson (1897)	Discovered the **electron**. "Plum pudding" model: a ball of **positive charge** with negative electrons embedded in it.	Cathode-ray experiments showed atoms contain much smaller, negatively charged particles — so atoms are *not* indivisible. Atoms are neutral, so there must be positive charge to balance the electrons.
Hans Geiger & Ernest Marsden (1909, working with Rutherford)	Alpha-particle scattering experiment: fired positive alpha particles at a very thin sheet of gold foil.	Most alpha particles passed straight through; a few were deflected; a very small number (about 1 in 8000) bounced back. This could not happen with the plum pudding model, where the positive charge is spread out.
Ernest Rutherford (1911)	**Nuclear model**: almost all the mass and all the positive charge is concentrated in a tiny central **nucleus**; electrons orbit around it; the atom is mostly empty space.	Explained the scattering results (see below).
Niels Bohr (1913)	Electrons orbit the nucleus in **fixed shells (energy levels)** at specific distances. Electrons can move between shells by absorbing or emitting energy.	Rutherford's model could not explain why orbiting electrons do not spiral into the nucleus, or why elements give out light of specific colours (line spectra). Bohr's calculations agreed with experimental observations.
James Chadwick (1932)	Discovered the **neutron** in the nucleus.	Explained why the mass of nuclei was bigger than the mass of their protons alone, and explained isotopes.

The Alpha-Particle Scattering Experiment

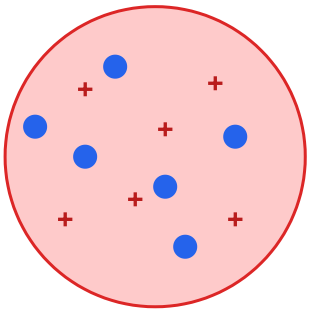
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Alpha Particles Passing Through a Layer of Gold Atoms

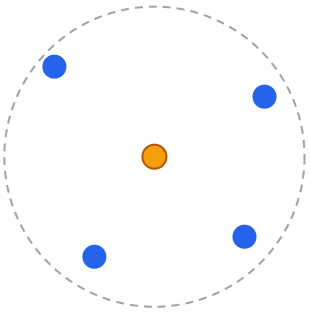


Observation	Conclusion
Most alpha particles passed **straight through** the foil	The atom is mostly **empty space**
Some were **deflected** through small angles	The centre of the atom is **positively charged** (it repels the positive alpha particles)
A very few **bounced back**	The positive charge and most of the **mass** are concentrated in a **very small** nucleus

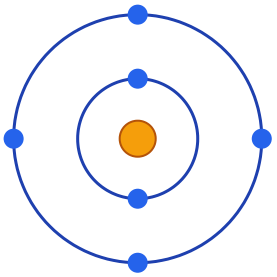
Three Models of the Atom



Thomson: plum pudding
electrons in a ball of positive charge



Rutherford: nuclear
tiny + nucleus; mostly empty space



Bohr: electron shells
electrons at fixed energy levels (carbon, 2,4)

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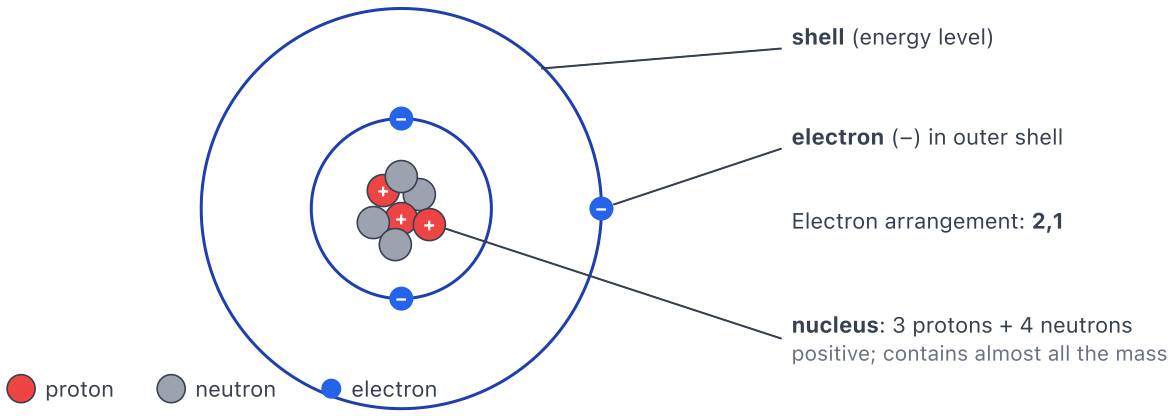
> **Exam Tip:** For "describe how the model changed and why" questions, always pair the **new evidence** with the **change it caused**. E.g. "Some alpha particles bounced back from the gold foil, which the plum pudding model could not explain, so Rutherford proposed that the positive charge and mass are concentrated in a tiny nucleus."

The Structure of the Atom Today

The modern model describes the atom as a **positively charged nucleus** surrounded by **negatively charged electrons**.

- The nucleus contains **protons** (positive) and **neutrons** (no charge).
- The **radius of the nucleus is much smaller than the radius of the atom** — about 1/10 000 (10^{-4}) of it.
- **Almost all of the mass** of the atom is in the nucleus, because protons and neutrons are about 1836 times heavier than electrons.
- The electrons occupy shells (energy levels) at different distances from the nucleus.
- An atom has **equal numbers of protons and electrons**, so it has **no overall charge**.

Structure of a Lithium-7 Atom (not to scale)



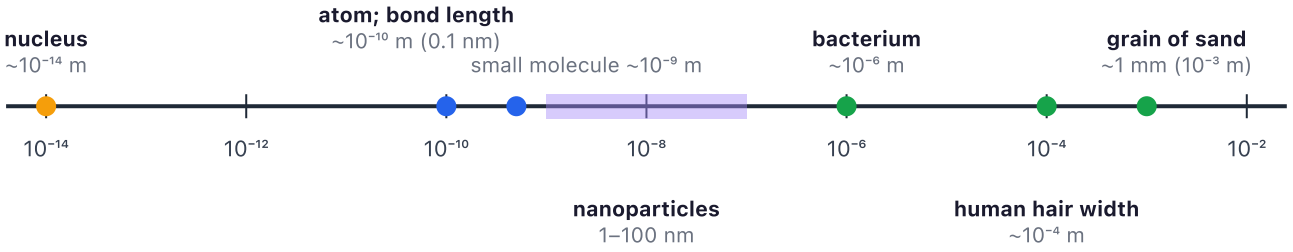
Size and Scale of Atoms

Atoms are extremely small. You need to recall the typical size (order of magnitude):

Typical atomic radius ≈ **1×10^{-10} m** (0.1 nm) · Typical bond length ≈ **10^{-10} m**

Radius of a nucleus ≈ 1×10^{-14} m (about 1/10 000 of the atomic radius)

Orders of Magnitude (metres)



Each step on the scale is ×100. 1 nm = 1 × 10^{-9} m.

Some ways to relate atomic sizes to the everyday world:

- A human hair is about 1 × 10^{-4} m wide — roughly **1 million carbon atoms** side by side ($10^{-4} \div 10^{-10} = 10^{6}$).
- If an atom were the size of a football stadium, the nucleus would be about the size of a pea in the centre.
- A 1 cm line of atoms would contain about 1 × 10^{8} atoms ($10^{-2} \div 10^{-10}$).

Worked Example: A gold atom has a radius of 1.4 × 10^{-10} m. Its nucleus has a radius of 7 × 10^{-15} m. How many times bigger is the atom than the nucleus?
1.4 × 10^{-10} ÷ 7 × 10^{-15} = 0.2 × 10^{5} = **2 × 10^{4} (20 000 times)**.

Worked Example: A nanoparticle has a diameter of 20 nm. About how many atoms (diameter 0.2 nm) would fit across it?
20 nm ÷ 0.2 nm = **100 atoms** — so nanoparticles contain a few hundred to a few thousand atoms in total (see C2.3).

Protons, Neutrons and Electrons

Particle	Relative charge	Relative mass	Where found

Proton	+1	1	Nucleus
Neutron	0	1	Nucleus
Electron	−1	very small (about 1/1840, ≈ 0.0005)	Shells around the nucleus

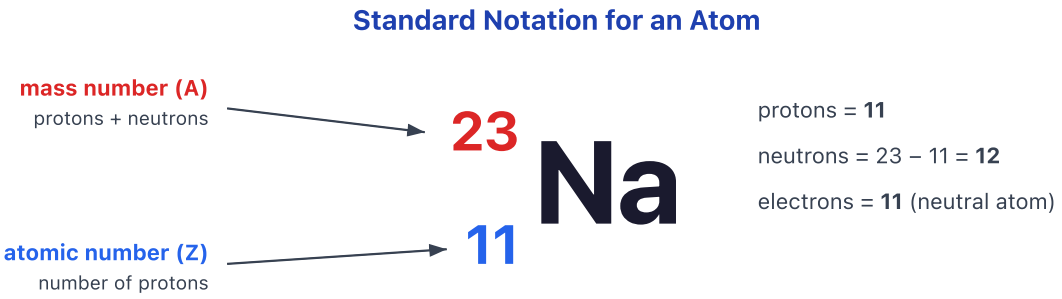
Exam Tip: Write the electron's relative mass as "very small", "negligible" or "1/1840" — never 0, and never 1. Relative charges must include the sign: +1, 0, −1.

Atomic Number, Mass Number and Standard Notation

Atomic number (Z) = number of **protons** in the nucleus. Every atom of the same element has the same atomic number, and it is different for every element. (It is also the number of electrons in a neutral atom.)

Mass number (A) = total number of **protons + neutrons** in the nucleus.

Number of neutrons = mass number − atomic number = **A − Z**.



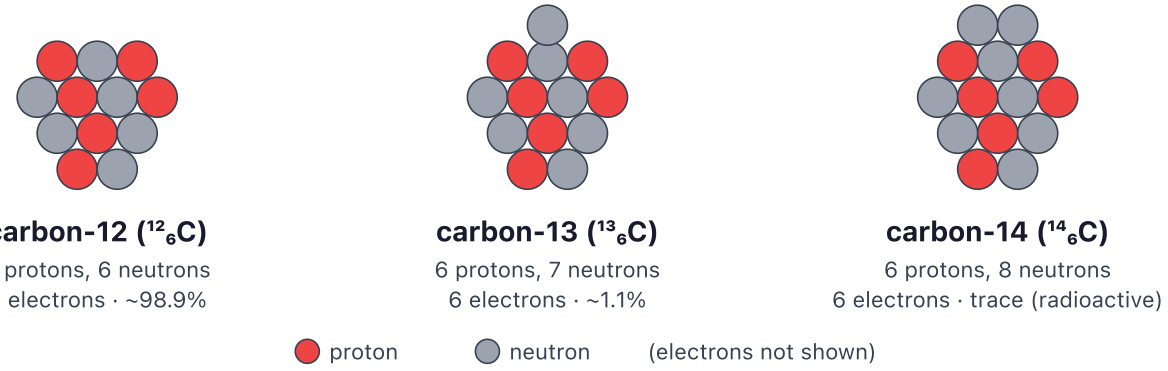
Mass number on top (bigger number), atomic number at the bottom.

On the Periodic Table, the larger number given for each element is usually the **relative atomic mass** (an average over its isotopes, C2.1) and the smaller number is the atomic number. For chlorine, A_r = 35.5 — not a whole number, because chlorine is a mixture of isotopes.

Isotopes

Isotopes are atoms of the **same element** (same number of protons) with **different numbers of neutrons**, and therefore different mass numbers.

Isotopes of Carbon: Nuclei Compared



- Isotopes have the **same chemical properties**, because chemical reactions depend on electrons, and isotopes have the same number of electrons (and the same electron arrangement).
- They can have slightly different **physical properties** (e.g. density) because their masses are different. Some isotopes are radioactive (e.g. carbon-14).
- Chlorine has two common isotopes: chlorine-35 (17 p, 18 n) and chlorine-37 (17 p, 20 n).
- Hydrogen has three: ^{1}H (no neutrons), ^{2}H deuterium (1 neutron), ^{3}H tritium (2 neutrons).

Exam Tip: A common misconception is that a neutral atom has equal numbers of protons and neutrons. It has equal numbers of **protons and electrons**. The number of neutrons can be anything (sodium-23 has 11 protons but 12 neutrons; uranium-238 has 92 protons and 146 neutrons).

Ions

An **ion** is a charged particle formed when an atom (or group of atoms) **loses or gains electrons**.

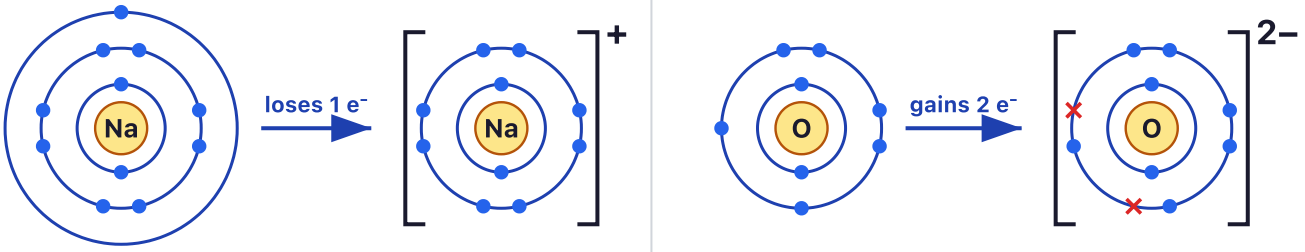
• Losing electrons → more protons than electrons → **positive ion** (e.g. Na^{+}, Mg^{2+}, Al^{3+}).

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- Gaining electrons → more electrons than protons → **negative ion** (e.g. Cl^-, O^{2-}, N^{3-}).

The **nucleus does not change** — the number of protons (and neutrons) stays the same.

Forming Ions: Electrons Are Lost or Gained



Na atom 2,8,1
11 p, 11 e^- → charge 0

Na^+ ion $[2,8]^+$
11 p, 10 e^- → charge +1

O atom 2,6
8 p, 8 e^- → charge 0

O^{2-} ion $[2,8]^{2-}$
8 p, 10 e^- → charge −2

Crosses (×) show the electrons gained. The nucleus (protons) never changes.

Calculating Protons, Neutrons and Electrons

$$\textbf{protons} = Z \quad \cdot \quad \textbf{neutrons} = A - Z \quad \cdot \quad \textbf{electrons} = Z - (\textbf{charge})$$

(for a 2+ ion subtract 2 electrons; for a 2− ion add 2 electrons)

Species	Atomic number	Mass number	Protons	Neutrons	Electrons
$^{19}_{9}F$ atom	9	19	9	10	9
$^{19}_{9}F^-$ ion	9	19	9	10	**10**
$^{24}_{12}Mg$ atom	12	24	12	12	12
$^{24}_{12}Mg^{2+}$ ion	12	24	12	12	**10**
$^{27}_{13}Al^{3+}$ ion	13	27	13	14	**10**
$^{16}_{8}O^{2-}$ ion	8	16	8	8	**10**
$^{37}_{17}Cl^-$ ion	17	37	17	20	**18**
$^{39}_{19}K^+$ ion	19	39	19	20	**18**

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Worked Example: An ion has 16 protons, 18 neutrons and 18 electrons. Identify it and write it in standard notation.
16 protons → atomic number 16 → **sulfur**. Mass number = 16 + 18 = 34. Electrons – protons = 18 – 16 = 2 extra electrons, so the charge is 2–.
Answer: $^{34}_{16}S^{2-}$ (a sulfide ion of the isotope sulfur-34).

Worked Example: Two particles, X and Y, both have 20 protons. X has 20 neutrons and 20 electrons; Y has 22 neutrons and 18 electrons. Explain how X and Y are related.
Both have 20 protons, so both are **calcium**. They have different numbers of neutrons, so they are **different isotopes** (calcium-40 and calcium-42). Y has 2 fewer electrons than protons, so Y is a **Ca^{2+} ion**, while X is a neutral atom.

Exam Tip: A positive ion has **lost electrons**, not gained protons. In "explain" answers, say e.g. "a magnesium atom loses two electrons to form Mg^{2+}; it still has 12 protons but only 10 electrons, so the overall charge is 2+".

Quick Summary — Key Ideas for C1

Idea	Key point
States of matter	Solid: regular, vibrate. Liquid: random, close, move past each other. Gas: random, far apart, fast.
Changes of state	Physical and reversible; mass conserved; temperature constant during the change.
Physical vs chemical change	Physical: same particles rearranged. Chemical: atoms rearranged into new substances.
(HT) Limitations of particle model	Ignores forces between particles, particle size and the space between particles.

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Atomic models	Dalton → Thomson (electron, plum pudding) → Geiger & Marsden / Rutherford (nucleus) → Bohr (shells).
Size	Atom radius & bond length ~10^{-10} m; nucleus ~1/10 000 of the atom.
Sub-atomic particles	p: +1, 1 · n: 0, 1 · e: –1, ~1/1840
Numbers	Z = protons; A = protons + neutrons; neutrons = A – Z; electrons = Z – charge
Isotopes	Same protons, different neutrons; same chemical properties
Ions	Formed by losing / gaining electrons; nucleus unchanged

Specification Checklist

Code	You should be able to...	Tier
C1.1a	Describe the main features of the particle model in terms of states of matter and change of state	F & H
C1.1b	Explain, using the particle model, the difference between physical and chemical changes	F & H
C1.1c	Explain the limitations of the particle model (inelastic spheres) in relation to changes of state: no forces, particle size or space between particles	**HT only**
C1.2a	Describe how and why the atomic model has changed (Dalton, Thomson, Rutherford, Bohr, Geiger and Marsden)	F & H
C1.2b	Describe the atom as a positive nucleus surrounded by negative electrons; nucleus tiny but contains most of the mass	F & H
C1.2c	Recall the typical size of atoms and small molecules (~10^{-10} m)	F & H
C1.2d	Recall relative charges and approximate relative masses of protons, neutrons and electrons	F & H

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C1.2e	Calculate protons, neutrons and electrons in atoms and ions from atomic and mass numbers; define ion, atomic number, mass number, isotope; use standard notation	F & H

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